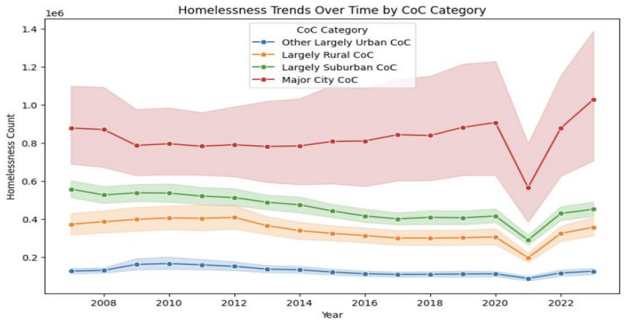
Predictive Model Based on Homelessness

PROJECT SUMMARY

This project aims to develop a predictive model of homelessness across 400 Continuums of Care in the U.S., which annually report homelessness counts. Using data from 2007-2023, the project explores the influence of structural factors such as rent, poverty, and other socioeconomic conditions on homelessness rates. It seeks to forecast homelessness trends and identify key community-level predictors.

HOMELESSNESS INSIGHTS



PROJECT RESULTS

We performed Random Forest and cross validation and tracked RMSE and R squared evaluation metrics for homelessness types under consideration.

TYPE (PER CAPITA)	RMSE	R^2
Overall	0.00099	0.7866
Individuals	0.00077	0.7802
Families	0.00038	0.8177
Unsheltered	0.00078	0.7001
Sheltered	0.00048	0.8798

CONCLUSION

Our project shows that Renter Household Rate, Total Population, and Total Beds are most influential variables. This demonstrates community level housing, socioeconomic conditions are essential in predicting homelessness trends.

PROJECT QUESTIONS

To what extent community-level measures of rent, poverty, and other housing market conditions accurately predict the number of persons experiencing homelessness in a community?

What community-level measures are most important in predicting the number of persons experiencing homelessness in a community?

TEAM MEMBERS

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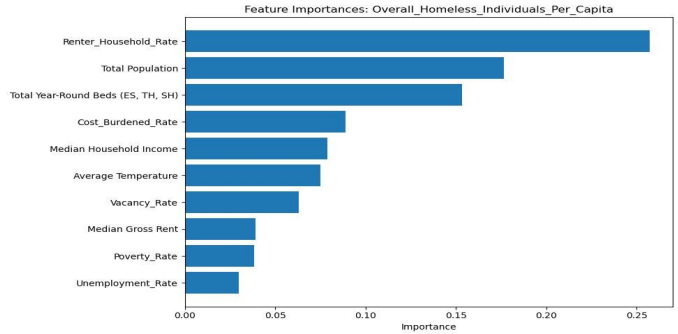
REFLECTIONS

Learning from Interdisciplinary Collaboration
Value of Data Preprocessing
Importance of Clear Communication and Iteration



CLIENTS & MENTORS

Clients: Dr. Tom Byrne, Dr. Molly Richard
EIR: Tom Hopper
TPM: Dhruv Shah
PM: Jasmine Dong

TECH STACK

Languages & Libraries: Python, R, Pandas, Scikit-learn, dplyr, tidyr